

Backend Developer Technical Test: Building a Railway Signals API

1. Objective

Develop a RESTful API in Go for managing Railway Signals. You will be provided with a JSON file containing relevant, but potentially quirky and unstructured data related to railway signals and tracks. Based on this data, you will design and implement the API, including defining endpoints, creating data models, and ensuring robust error handling and data validation. Using this API create a React web app using Vite, TailwindCSS, Tanstack Query and Zod to validate and present the data.

JSON File

<https://www.dropbox.com/scl/fi/e6umx06zfxlijwqyf30oq/crosstech-test-data.json?rlkey=fg8v7mob1ndjiv3pc3f35jvbs&dl=0>

Tech Stack:

- GoLang / Typescript (React)
- Any Framework (Preferable Echo)

2. Requirements

2.1. Data Modeling

- Analyze the provided JSON data and design suitable Go structs to represent the entities and their relationships. Note: Data was extracted using [Python](#), so look out for any awkward mismatches in data.
- Consider data normalization, efficiency, and maintainability when designing the data model.
- No need to save it in database, local memory will work or using json as database

2.2. API Design and Implementation

- Based on your data model, design and implement a RESTful API (GET) to manage the railway data.
- **Required Endpoints:**
 - Relevant endpoint to manage signals
 - Relevant endpoint to manage tracks
- **Optional Endpoints (Bonuses):**
 - **Signal-Track Relationship:** Retrieve all tracks associated with a signal

2.3. Frontend

Requirements

- Create React app using Vite with TailwindCSS as the styling solution
- Query the API using Tanstack Query
- Validate the API response using the Zod validation library
- Display the track and signal data appropriately
- Clearly visualise how tracks and signals relate to each other
- Implement filtering for tracks by track_id and source/target name

What we're evaluating

- Proper use of TanStack Query (queries, loading states, error handling)
- Proper React patterns, hooks usage and organization
- Effective use of Tailwind CSS
- How to best visualize the model relationships (list, table, card etc.)
- Code clarity and organization

Optional Objectives

- Map view to display locations of tracks and signals (react-map/react-leaflet can help with this)
- Pagination for the track and signal views

2.4. Core Implementation Details

- **Code Quality:** Adhere to Go/Typescript coding conventions and best practices. Write clean, well-documented, and maintainable code.
- **Error Handling:** Implement proper error handling throughout the application. Return informative error messages in API responses.
- **Data Validation and Integrity:** Validate input data and handle data inconsistencies and provide clear error messages or warnings.
- **Comments and Documentation:** Add clear and concise comments to your code to explain any design decisions, complex logic, or non-obvious choices. Provide documentation for the API, including descriptions of the endpoints, request/response formats.

3. Submission

1. **Create a Git Repository:** Create a private Git repository on a platform like GitHub, GitLab, or Bitbucket. Initialize the repository with a README file and any other necessary files (e.g., .gitignore).
2. **Develop the Solution:** Develop your solution in the repository, committing your changes regularly with clear and concise commit messages.
3. **Submit the Repository Link:** Once you have completed the test, submit the link to your

Git repository via email to harry@crosstech.co.uk and invite harryevans1997 and MaxG201 as collaborators

5. Additional Notes

- This test is designed to take no more than 2-4 hours to complete.
- A sample JSON file with relevant data will be provided.
- **If you encounter any issues or have any questions during the test, please reach out to harry@crosstech.co.uk for clarification or assistance.**
- **We are not expecting perfection in this test. We are primarily interested in seeing how you approach the problem, your coding style, and your ability to think critically and solve problems.**